UNIVERSITY OF PERADENIYA
DEPARTMENT OF PHYSICS

END SEMESTER EXAMINATION
SEMESTER II - 2022/2023
PHY1203/PH 102 - General Physics II

S/21/[redacted]

Time allowed: ***Three hours***

Answer **Five** questions selecting **at least Two from section-A, at least Two from section-B** and **only One from section-C**
(Use of calculators is allowed)
Symbols have their usual meanings unless specified otherwise

Electron charge (e) $= 1.60\times10^{-19}$ C
Mass of electron (m_e) $= 9.11\times10^{-31}$ kg
Mass of proton (m_p) $= 1.67\times10^{-27}$ kg
Permittivity of free space (ε_0) $= 8.85\times10^{-12}$ F m^{-1}
Permeability of free space (μ_0) $= 4\pi\times10^{-7}$ H m^{-1}

Planck's constant (h) $= 6.63\times10^{-34}$ J s
Speed of light (c) $= 3.00\times10^{8}$ m s^{-1}
Wien's constant $= 2.89\times10^{-3}$ m K
Stefan-Boltzmann constant (σ) $= 5.67\times10^{-8}$ W m^{-2} K^{-4}
Avogadro number (N_A) $= 6.02\times10^{23}$ mol^{-1}

Section-A

01. (a) Write down Coulomb's law in electrostatics in words and give it as an expression.

(b) Give the relationship between the electric field intensity $E(r)$ and the potential $V(r)$.

(c) As shown in Figure 1, a conducting rod AB of length L and negligible cross section carries a charge Q uniformly distributed along its length. To find the electric field at point P, located a distance a from end B, consider a small element of length dx of the rod at a distance x from P.

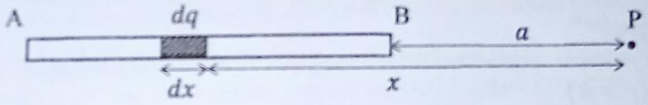


Figure 1

(i) Show that the electric field (dE) at point P due to the element charge dq on the rod is given by: $dE = \frac{\lambda dx}{4\pi\varepsilon_0 x^2}$, where λ is the linear charge density of the rod.

(ii) Obtain an expression for the electric field E at point P due to the entire charge distribution on the rod.

(iii) Show that the electric potential at point P due the above charge distribution is given by: $V = \frac{Q}{4\pi\varepsilon_0 L} ln\left(\frac{L+a}{a}\right)$.

02. (a) (i) Clearly identifying the quantities involved, state Ampere's Circuit Law in vector form.

(ii) Using Ampere's Circuit Law, derive an expression for the magnetic field intensity at a point near infinitely long, straight conductor carrying a steady current.

(b) A long, straight, cylindrical conducting wire of radius R carries a current such that the current density (J) is proportional to r , where r ($r < R$) is the distance from the central axis of the wire [see Figure 2 (a) and 2 (b)]

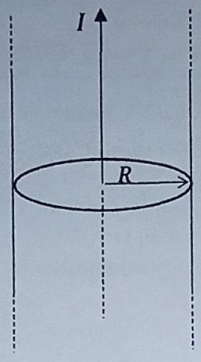


Figure 2 (a)

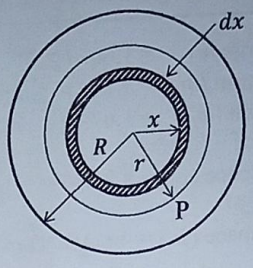


Figure 2 (b)

Consider a point P inside the wire at a distance r from the axis [see Figure 2 (b)]. Assume $J = kr$, where k is a constant, to find the current enclosed by the cross-sectional area with radius r.

(i) Show that the current through the elemental ring of radius x and thickness dx as shown in Figure 2 (b) is given by $di = 2\pi kx^2 dx$.

(ii) Use the above result to show that the current enclosed by the cross-sectional area of radius r is $i_{enclosed} = \frac{2}{3}\pi kr^3$.

(iii) If the total current through the wire is I, show that: $k = \frac{3I}{2\pi R^3}$.

(iv) Use Ampere's Circuit Law to show that the magnetic field intensity at P is: $B = \frac{1}{3}\mu_0 kr^2$.

(v) Obtain an expression for the magnetic field intensity at a point outside the wire ($r > R$).

(vi) Plot the variation of magnetic field intensity (B) with the distance (r) from the centre axis of the wire.

03. (a) (i) What is the Lorentz force? Explain the factors that determine its magnitude and direction.

(ii) The Lorentz force does not do work on a charged particle moving in a magnetic field. Explain why?

(b) A charged particle of mass m and charge q enters a magnetic field of intensity B, as shown in Figure 3. Particle exits the magnetic field without any change in its speed as illustrated.

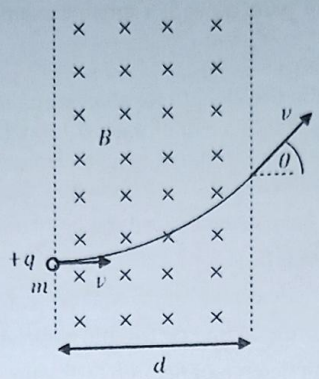


Figure 3

(i) Obtain an expression for the radius of curvature R of the particle's path in terms of m, v, B and q.

(ii) If the magnetic field is limited to a region of width d, find the angle θ and the time t spent inside the field region for: (1) $d = \frac{mv}{2qB}$, (2) $d = \frac{mv}{qB}$ and (3) $d = \frac{2mv}{qB}$

(c) A charged particle with charge q and mass m enters a region having uniform magnetic field B and electric field E with an initial velocity of v_0.

(i) Explain, if the electric and magnetic fields are parallel to each other, how would the particle's trajectory be changed? Discuss the nature of the motion in this case.

(ii) Explain, if the electric and magnetic fields are perpendicular to each other, what type of motion would the particle follow? Explain how both fields contribute to the motion.

Section-B

04. (a) (i) What is Compton scattering?
(ii) Write down the formula for the Compton shift, identifying all the symbols and stating all the important assumptions.
(iii) With the help of diagrams, explain the results of the Compton scattering experiment.
(iv) Why do we need high energy radiation such as X-rays or γ-rays to observe the Compton scattering?
(v) Justify your answer for part (iv) with a suitable example.

(b) (i) State de Broglie hypothesis.
(ii) Compare the de Broglie wavelength for an electron moving at a speed equal to 1.00×10^7 m s^{-1} with that of a ball of mass 0.150 kg travelling at 50.0 m s^{-1}.
(iii) Compare these wavelengths with that of an electron traveling at 0.999c.
(iv) The neutrons produced in a reactor are known as thermal neutrons, because their kinetic energies have been reduced (by collisions) to $\frac{3}{2}kT$, where T is room temperature (293 K).
I. What is the kinetic energy of thermal neutrons?
II. What is the corresponding de Broglie wavelength?
III. Describe an application of neutron diffraction and justify the reasons for its use?

05. (a) (i) State Wien's Displacement Law and the Stefan-Boltzmann Law. Clearly define all the symbols used in these laws.
(ii) At what wavelength does an object emit the maximum thermal radiation at room-temperature (say $T = 20$ °C)?
(iii) To what temperature must we heat the object until its peak thermal radiation is in the red region of the spectrum (say $\lambda = 650$ nm)?
(iv) What is the ratio between the total intensity at the two temperatures in part (ii) and (iii)?

(b) (i) A laser beam with an intensity of 120 W m^{-2} is incident on a surface of sodium sample. It takes a minimum energy of 2.3 eV to release an electron from sodium sample. Using classical ideas and assuming electron to be confined to an area of radius equal to that of a sodium atom (0.10 nm), calculate how long will it take for the surface to absorb enough energy to release an electron?
(ii) What might be the observations of the photoelectric effect experiment if we
(1) double the frequency of the incident light?
(2) double the wavelength?
(3) double the intensity?
(iii) Photoelectrons released from zinc by ultraviolet light were stopped by a voltage of 4.3 V. Find K_{max} and v_{max} for these electrons.

06. (a) (i) Using the Bohr's model, derive an expression for the electron energy of the nth level of a Hydrogen like atom with a nucleus of charge Ze.

(ii) Hence derive an expression for the wavelength of a photon emitted for the electron transfer from the energy level n_i to n_f.

(iii) an American astronomer Edward Charles Pickering in 1896 observed that lines in a spectrum of star fits the following empirical formula

$$\frac{1}{\lambda} = R\left[\frac{1}{\left(\frac{n_f}{2}\right)^2} - \frac{1}{\left(\frac{n_i}{2}\right)^2}\right]$$

where R is the Rydberg constant. Show that these lines can be explained by the Bohr theory as originating from He^{+2}.

(b) (i) Show that the law of radioactive decay is given by $N = N_0 e^{-\lambda t}$.

(ii) Define the half-life $T_{1/2}$ and obtain an expression for $T_{1/2}$.

(iii) The half-life of the radioactive nucleus $^{226}_{88}Ra$ is 1.6×10^3 years.

I What is the decay constant of $^{226}_{88}Ra$?

II If a sample contains 3.0×10^{16} radioactive nuclei, determine its activity.

III What is the decay rate of the sample after 2.0×10^3 years?

Section-C

07. (a) Aluminum and Silicon are widely used in modern electronic and optoelectronic devices. Aluminum has a low melting point, low stiffness (effective spring constant) which is half that of Silicon and a high thermal expansion coefficient which is twice the value of Silicon. Qualitatively sketch the interatomic potential vs separation curves for Aluminum and Silicon in the **same diagram.**

(b) Potential energy of a diatomic molecule is given by the Lennard–Jones potential,

$$V(r) = \frac{A}{r^{12}} - \frac{B}{r^6}$$

where A and B are constants and r is the separation between the atoms.

For an N_2 molecule, $A = 0.755 \times 10^{-119}$ eV m^{12} and $B = 0.853 \times 10^{-59}$ eV m^6.

(i) Identify the attractive potential and the repulsive potential and comment on their origin.

(ii) Find the separation r_0 at which the energy of the molecule is a minimum.

(iii) Find the minimum energy E_0 required to break up the N_2 molecule.

(iv) Find the effective spring constant, k of the N_2 molecule.

08. In an HCl molecule, take the Cl atom to be the isotope ^{35}Cl. The equilibrium separation of the H and Cl atoms is 0.12746 nm. The atomic mass of the H atom is 1.007825 u and that of the ^{35}Cl atom is 34.968853 u. $1\text{u} = 1.66 \times 10^{-27}\text{kg}$.

(***Hint: Find all required quantities in your calculations up to 3 decimal places***)

(a) Calculate the longest wavelength in the rotational spectrum of this molecule.

(b) Repeat the calculation in part (a), but take the Cl atom to be the isotope ^{37}Cl, which has atomic mass 36.965903 u. The equilibrium separation is the same as in part (a).

(c) Naturally occurring chlorine contains approximately three parts of ^{35}Cl to one part of ^{37}Cl.

(i) Each line in the microwave rotational spectrum of HCl splits into a doublet. Explain why?

(ii) Calculate the separation in wavelength between the doublet lines for the longest wavelength.

(d) Now, consider only the vibrational motion of the molecule HCl with isotope ^{35}Cl. Find the longest wavelength of the vibrational spectrum of this molecule if the effective spring constant of the molecule, $k = 480\ \text{N m}^{-1}$.

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